

Securing data of biotechnological laboratories using blockchain technology

Krzysztof Misztal^{1,2}
Tomasz Służalec^{1,2} Aleksandra Kubica-Misztal²

Jagiellonian University, Poland
{krzysztof.misztal,t.sluzalec}@uj.edu.pl

diCELLa Ltd, Poland
a.kubica-misztal@dicella.com

19th International Conference on Computer Information Systems and
Industrial Management Applications, 2020

Overview

- 1 Introduction
- 2 Sensitive data in blockchain
- 3 Laboratories
- 4 Technological details
- 5 Blocks implementation details
- 6 System details
- 7 Platform
- 8 System stability
- 9 Conclusions

Introduction

Bitcoin

A few years ago blockchain technology was used in cryptocurrency. Nowadays, a variety of diverse areas are seeing the benefits of applying this technological approach to their needs.

Blockchain system. In 2008, Satoshi Nakamoto published a paper [Nakamoto, 2008] describing bitcoin, which is now referred to as the first cryptocurrency in the world.

It is worth noticing that while all bitcoin transactions are stored in the blockchain unencrypted, it is possible to keep encrypted data in the blockchain.

Introduction

Bitcoin transactions

Bitcoin scenario:

- 1 John wishes to pay bitcoins to Alice.
- 2 John's transaction is made.
- 3 Transactions are combined into a new block.
- 4 Blocks are broadcasted to miners.
- 5 Miners validates and accepts the transactions.
- 6 New block is encrypted with hash of last block.
- 7 New block become a part of blockchain.
- 8 Alice receives bitcoins.

Sensitive data in blockchain

Centralized system vs decentralized system

The centralized system is vulnerable to hacker attacks, as well as phishing. The solution is for the user to own and be responsible for providing its information. We can do this by applying blockchain storage solution proposed by [Zyskind, 2015].

Sensitive data in blockchain

Data access permissions

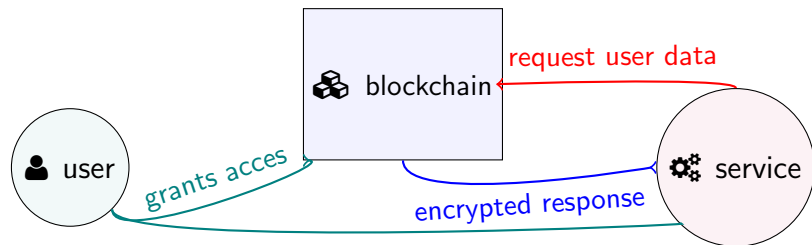


Figure: Simplified schema of the above grant-access data blockchain.

Some ideas about blockchain application in health care can be seen in [Angraal, 2017].

Laborator data objectives

The results obtained in GMP and GLP laboratories have to be reliable. Any modification of partial results can make the final result fail. It is the reason why the lab records have to be collected precisely and without any forgery. It is also important to remember every the author of every record. The authorship has to be beyond any doubt, which means that nobody can create a record on behalf of another user and no user can deny his own authorship of any of his records. Moreover, the documentation has to be immutable so any stored data cannot be modified nor deleted. These requirements make users keep exact records, which objectively increases the reliability of the laboratory.

Technological details

Programming language and packages

We choose specific cryptographic packages and programming language Python3 so we can create application on any technology. Although python apps are not as efficient as in C language but they are more portable. What is more professional programmers that are working with cyber-security chooses also python. In the prototype we used Django framework and cryptography python package because of the SHA-3 implemented hash function. It allows to simple and effective programming and is extending rapidly. After the theoretical analysis ensures us that these algorithms are sufficient to our needs.

Technological details

Blocks security

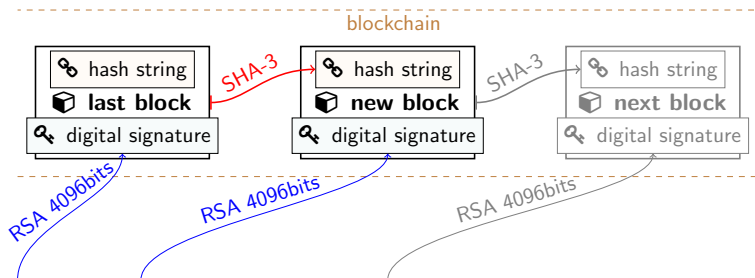


Figure: Hash linking and RSA signature in blocks.

Blocks implementation details

Block fields

To avoid the future problems we designed the blocks in the form of dictionary with a predefined set of keys. Those objects can be easily serialized and deserialized to the JSON format.

```
class BlockField(Enum):  
    BLOCK_TYPE = 0  
    TIMESTAMP = 1  
    AUTHOR = 2  
    SIGNATURE = 3  
    OBJECT = 4  
    PREVIOUS_HASH = 5  
    NONCE = 6
```

Listing 1: Enum describing all parameters in block object.

Blocks implementation details

Block types

We also defined five types of blocks:

```
class BlockType(Enum):  
    GENESIS = 0  
    INSERT_BLOCK = 1  
    UPDATE_BLOCK = 2  
    INSERT_KEY = 3  
    REVOKE_KEY = 4
```

Listing 2: Enum describing possible types of blocks in the chain.

Blocks implementation details

Blockchain performance

Remark

We have confirmed the effectiveness in every aspect of the use cases, as well as in the performance of our system. The processing time and memory usage by blockchain mechanism were not computation demanding.

System details

Genesis block

The *genesis block* is a first one in the chain. There can be only one *genesis block* in the blockchain. It contains the system administrator key and it cannot be rejected. Public key is controlled by admins managing the software, not in the hands of laboratory employees.

Genesisblock technical details:

- 1 All user information are in blockchain.
- 2 In UserData field there are private user keys encrypted by their passwords.
- 3 There is only one superuser admin.
- 4 Only admin is permitted to create genesis block.
- 5 There is only one genesis block that starts the blockchain.

Users technical details:

- Genesis block is mandatory to start blockchain and create new user.
- User permission and private keys information are checked every time.
- To secure unauthorized accessing permissions are stored in blockchain similar to admin and genesisblock scenario.
- Only superusers can add new authorized user.
- Only superusers can revoke permission to user.
- All actual user privileges are stored in summary block.

System details

Summary (short-cut) block

Omitting specific computational research units there might be lack of high-end computers in some laboratories. That is why we introduced summary-block idea. This simple addition is a perfect solution to shorten computation time while still not breaking blockchain basis.

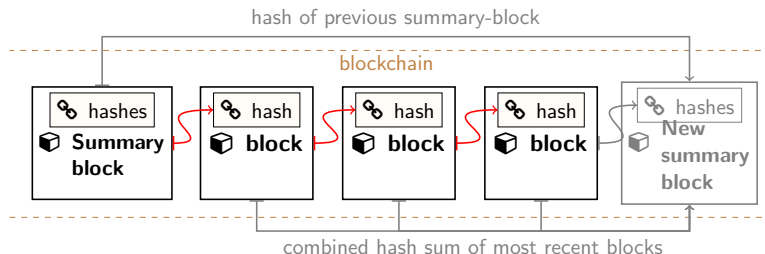


Figure: New summary-block contain K last blocks hashes and the hash of last summary-block. Summary block contains updated information of user privileges. Summary-block is a checkpoint block, there is no need to calculate "tail" of blockchain.

Platform

Usage example

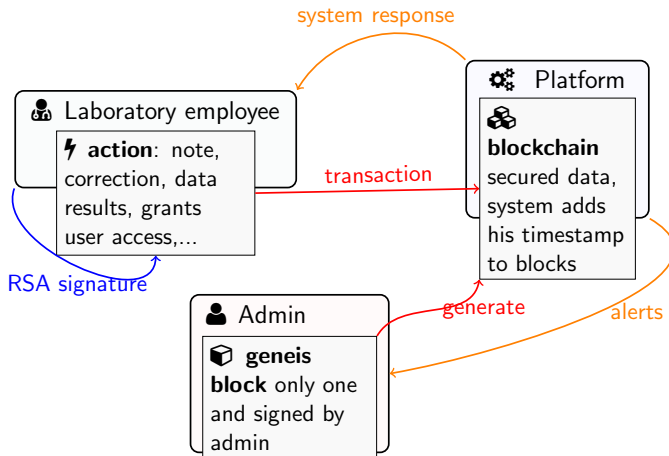


Figure: Platform is validating and storing signed and secured transaction. Additional lab-supporting algorithms perform analyses for the users basing on the blockchain data.

We validated time resources that were saved by our system. There was a significant improvement. System was effective to time-consuming and resource-consuming problems. The designed platform was verified in laboratory conditions. We observed that the designed solution had an impact on the laboratory workflow.

We have confirmed the effectiveness in every aspect of the use cases, as well as in the performance of our system. The processing time and memory usage by blockchain mechanism were not computation demanding.

The list of advantages that comes from proposed blockchain system:

- ① Data are secured against unauthorized changes.
- ② Blockchain security of data makes them authentic and repudiation.
- ③ Data stored in blockchain are transparent.
- ④ Due to GMP CFR 21 part 11 system allows to electronic documentation conducting.
- ⑤ System fulfils the acts of biobank laws.
- ⑥ System helps realization in tissue and cell procurement in order with doctor and dentist's law.
- ⑦ System fulfills the acts of pharmaceutic laws in terms of production advanced medicinal products.
- ⑧ System satisfies UE documents regarding production and medical therapy (transplanting cells, tissue and organs).
- ⑨ System additional algorithms archive the statistics improving the lab workflow.
- ⑩ System helps in stable managing the laboratory storage magazine.

System stability

New data

When the new data is added system is validating its propriety.

For example:

- if the note was made before,
- if edited content of the note corresponds to the last one in blockchain,
- if a user is allowed to add a note - system verifies if user *digital key* is active and permitted to add a content.

System stability

New block

Below is the checklist that is validated by system when inserting a block to blockchain:

- 1 the first block is not a *genesis block*,
- 2 more then one block is *genesis block*,
- 3 *prev_hash* filed for *genesis block* is different then expected,
- 4 *timestamp* from name of *block* is not corresponding to the content of block,
- 5 *prev_hash* value of the *block* is not compatible with its *previous block*,
- 6 *hash string* do not begins with fixed length of zeros,
- 7 user *digital key* is added twice,
- 8 user *digital key* is deleted twice,
- 9 *genesis key* is deleted,
- 10 block added by unauthorized user - due to no *digital key*,
- 11 block added by unauthorized user - due to *digital key* rejection,
- 12 lack of mandatory filed in *block*.

Conclusions

The blockchain technology is widely used in accounting and banking, but not exclusively in those fields. There are many projects adapting this technology in many different areas of daily lives. Our project, by using blockchain technology in the widely understood area of healthcare, is among them.

References



Angraal, Suveen and Krumholz, Harlan M and Schulz, Wade L (2017)

Blockchain technology: applications in health care

Circulation: Cardiovascular Quality and Outcomes.



Dworkin, Morris J (2015)

SHA-3 standard: Permutation-based hash and extendable-output functions

Federal Information Processing Standards(NIST FIPS) 202.



Gallagher, Patrick (2013)

Digital signature standard (DSS)

Federal Information Processing Standards Publications, volume FIPS 186–3.



National Institute of Standards and Technology (2012)

Secure Hash Standard

Federal Information Processing Standard (FIPS) 180-4.

References



Satoshi Nakamoto (2008)

Bitcoin: A peer-to-peer electronic cash system

Manubot.



Schneier, B (2013)

The NSA is breaking most encryption on the internet

Blog post.



Stevens, Marc and Bursztein, E and Karpman, P and Albertini, A and Markov, Y and Bianco, A Petit and Baisse, C (2017)

Announcing the first SHA1 collision

Google Security Blog.



Zyskind, Guy and Nathan, Oz and others (2015)

Decentralizing privacy: Using blockchain to protect personal data

2015 IEEE Security and Privacy Workshops 180–184.

Thank You